

Master's thesis title

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Master's Thesis submitted to obtain
the degree of Master of Science in
programme name specialisationprogramme
Engineering Technology

Academic year 2024 - 2025

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In the preface, you write a personal account with an acknowledgement or expression of gratitude to the people who assisted you in completing your master's thesis and supported you during your studies.



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Samenvatting

Write a short summary in Dutch that is accessible to a wide audience and contains a **maximum of 3500 characters**. This summary must also be uploaded to KU Locket.

Summary

This (short) summary, accessible to a wide audience, is written in English and contains a **maximum of 3500 characters**. This summary must also be uploaded to KU Locket.

Keywords: Enter (more or less) five keywords.

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Acronyms

IoT Internet of Things. 1

LPWAN low-power wide-area network. 1

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1 Changes w.r.t. the default template

1.1 Example text

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1.2 Template features

- Changed font from default computer modern to CMBright.
- Increased line spacing in the title to prevent letters from overlapping.
- The siunitx package is included; see examples at: <https://github.com/dramco-edu/LaTeX/wiki/LaTeX-Tips-and-Tricks#using-si-units-m>
- Include units alongside equations using the following command:

```
\begin{equation}
E = m c^2
\tagaddtext{\si{joule}}
\end{equation}
```

, which outputs:

$$E = mc^2 \quad [\text{J}] \quad (1.1)$$

- Abbreviations:
 - Internet of Things (IoT)
 - low-power wide-area networks (LPWANs)
 - Internet of Things
 - IoT
 - Internet of Things (IoT)
- `\ce{CO2}` CO₂
- Refs:
 - Figure 1.1
 - Table 1.1
 - Chapter 1
 - Equation (1.1)
 - `\cref{fig:example}` becomes Figure 1.1
 - `\cref{tab:example}` becomes Table 1.1
 - `\cref{ch:introduction}` becomes Chapter 1
 - Equation (1.1)
- numerical citation [1]



Figure 1.1: Example JPG

- `\cite{callebaut10lorawan}` becomes [2]
- `\citeauthor{callebaut10lorawan}` becomes Callebaut
- `\citeauthorref{callebaut10lorawan}` becomes Callebaut [2]
- `\autocite[chap.~2]{callebaut10lorawan}` becomes [2, chap. 2]
- `\footcite{callebaut10lorawan}` becomes¹
- `\fullcite{callebaut10lorawan}` becomes G. Callebaut, “Lorawan network simulator”, *doi*, vol. 10, p. 5281,

- A 45° angle or a 45°.
- It is 17°C outside.
- 10 000
- 3.45×10^{-4}
- kg m s^{-2}
- kg m/s^2
- 10 %
- 68 kg

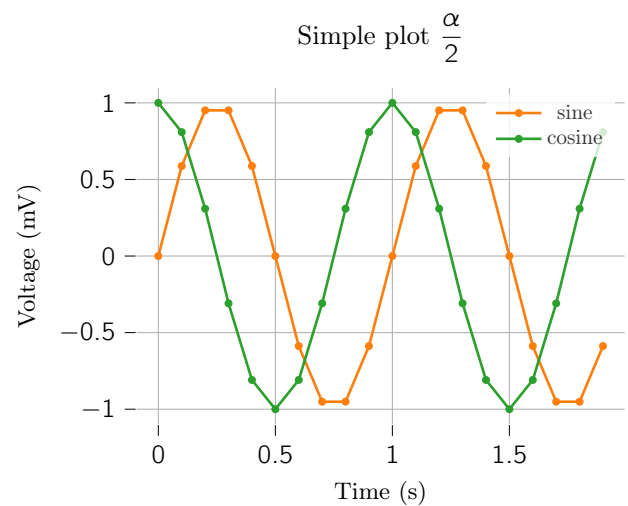


Figure 1.2: Caption

¹2.

Day	Min Temp	Max Temp	Summary
Monday	11C	22C	A clear day with lots of sunshine. However, the strong breeze will bring down the temperatures.
Tuesday	9C	19C	Cloudy with rain, across many northern regions. Clear spells across most of Scotland and Northern Ireland, but rain reaching the far northwest.
Wednesday	10C	21C	Rain will still linger for the morning. Conditions will improve by early afternoon and continue throughout the evening.

Day	Min Temp	Max Temp	Summary
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Wednesday	10C	21C	Rain will still linger for the morning. Conditions will improve by early afternoon and continue throughout the evening.

Table 1.1 Fixed-width columns.

	Treatment A	Treatment B
John Smith	Good response, no side-effects	No response
Jane Doe	–	Good response, no side-effects
Mary Johnson	No response	Good response with side-effects

```

import numpy as np

def incmatrix(genl1,genl2):
    m = len(genl1)
    n = len(genl2)
    M = None #to become the incidence matrix
    VT = np.zeros((n*m,1), int) #dummy variable

    #compute the bitwise xor matrix
    M1 = bitxormatrix(genl1)
    M2 = np.triu(bitxormatrix(genl2),1)

    for i in range(m-1):
        for j in range(i+1, m):
            [r,c] = np.where(M2 == M1[i,j])
            for k in range(len(r)):
                VT[(i)*n + r[k]] = 1;
                VT[(i)*n + c[k]] = 1;
                VT[(j)*n + r[k]] = 1;
                VT[(j)*n + c[k]] = 1;

            if M is None:
                M = np.copy(VT)
            else:
                M = np.concatenate((M, VT), 1)

            VT = np.zeros((n*m,1), int)

    return M

```

Listing 1: Minimal working example

Bibliography

- [1] G. Ottoy, G. Leenders, and G. Callebaut, "Lorawan efm32",
- [2] G. Callebaut, "Lorawan network simulator", *doi*, vol. 10, p. 5281,

Appendix A

An appendix

Add supporting material here.

Appendix B

Scientific article describing this thesis

The thesis should also contain a short scientific article. If you write your thesis in Dutch, you must write the article in English, and vice versa. We advise you to employ the IEEE Manuscript Templates for Conference Proceedings (<https://www.overleaf.com/latex/templates/ieee-conference-template/grfzhhnscsfqn>). Compile the article in another project and include the generated pdf file as shown below:

Conference Paper Title*

*Note: Sub-titles are not captured in Xplore and should not be used

1st Given Name Surname

dept. name of organization (of Aff.)

name of organization (of Aff.)

City, Country

email address or ORCID

2nd Given Name Surname

dept. name of organization (of Aff.)

name of organization (of Aff.)

City, Country

email address or ORCID

3rd Given Name Surname

dept. name of organization (of Aff.)

name of organization (of Aff.)

City, Country

email address or ORCID

4th Given Name Surname

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name of organization (of Aff.)

City, Country

email address or ORCID

5th Given Name Surname

dept. name of organization (of Aff.)

name of organization (of Aff.)

City, Country

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6th Given Name Surname

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Abstract—This document is a model and instructions for $\LaTeX$. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

This document is a model and instructions for $\LaTeX$. Please observe the conference page limits.

II. EASE OF USE

A. Maintaining the Integrity of the Specifications

The IEEEtran class file is used to format your paper and style the text. All margins, column widths, line spaces, and text fonts are prescribed; please do not alter them. You may note peculiarities. For example, the head margin measures proportionately more than is customary. This measurement and others are deliberate, using specifications that anticipate your paper as one part of the entire proceedings, and not as an independent document. Please do not revise any of the current designations.

III. PREPARE YOUR PAPER BEFORE STYLING

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections III-A–III-E below for more information on proofreading, spelling and grammar.

Keep your text and graphic files separate until after the text has been formatted and styled. Do not number text heads— $\LaTeX$ will do that for you.

Identify applicable funding agency here. If none, delete this.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: “Wb/m²” or “webers per square meter”, not “webers/m²”. Spell out units when they appear in text: “. . . a few henries”, not “. . . a few H”.
- Use a zero before decimal points: “0.25”, not “.25”. Use “cm³”, not “cc”.)

C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \quad (1)$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not

Appendix C

Poster

Appendix D

GenAI code of conduct

Placeholder

GenAI code of conduct

Download the most recent KU Leuven GenAI code of conduct, fill it in, export the completed document to PDF, and manually include that PDF here.

The files `chapters/GenAI_Code of conduct_AJ 2526_*` are provided as a starting point only. Always check the KU Leuven website for the most recent version before submitting your thesis.

Replace this placeholder with, for example:

```
\includepdf [pages=-]{chapters/code-of-conduct.pdf}
```